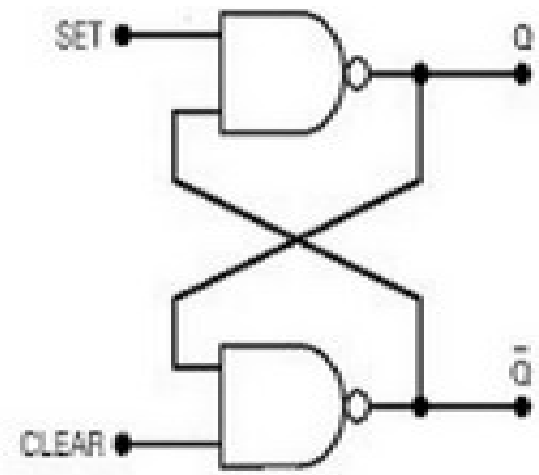
Latch SR (Flip-flop SR)

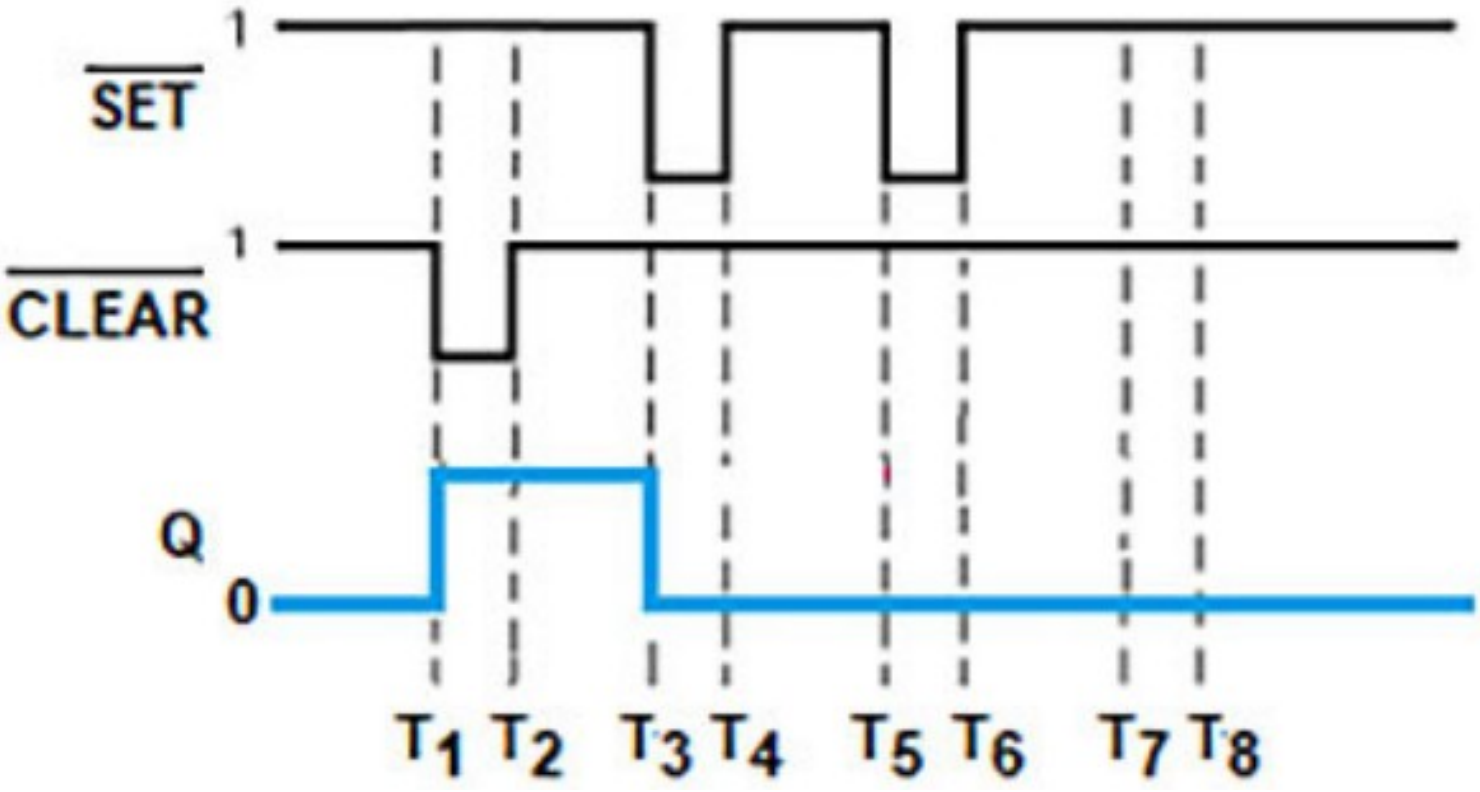
Permite dividir frequências, deslocar palavras, armazenar 1 bit



SET (S)	CLEAR (RESET) (R)	Q	$\overline{Q}$
0	0	Condição proibida	
0	1	0	1
1	0	1	0
1	1	Não muda (mantém a condição anterior)	

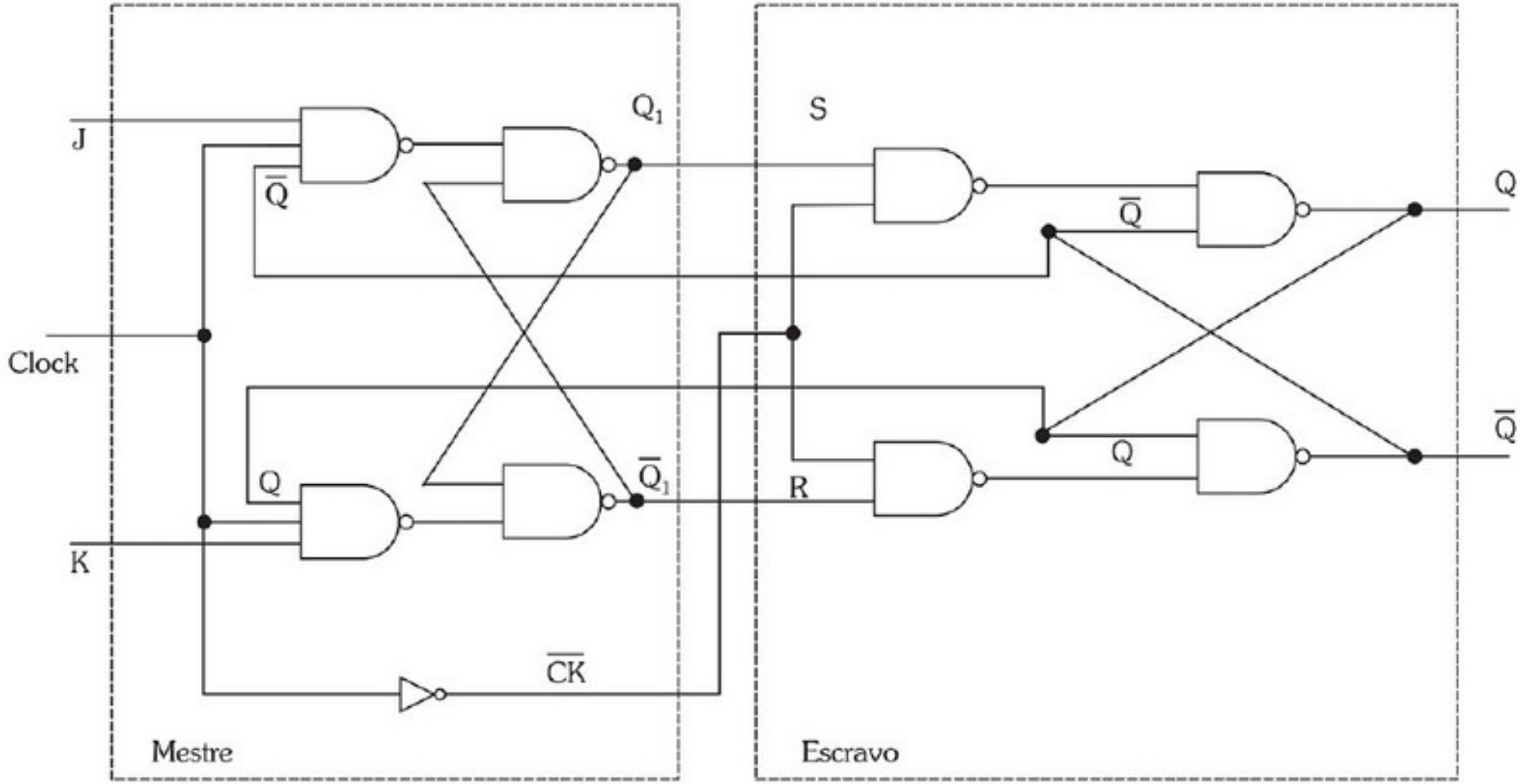
Flip-flop RS

Apresenta condição proibida quando S e R são 0

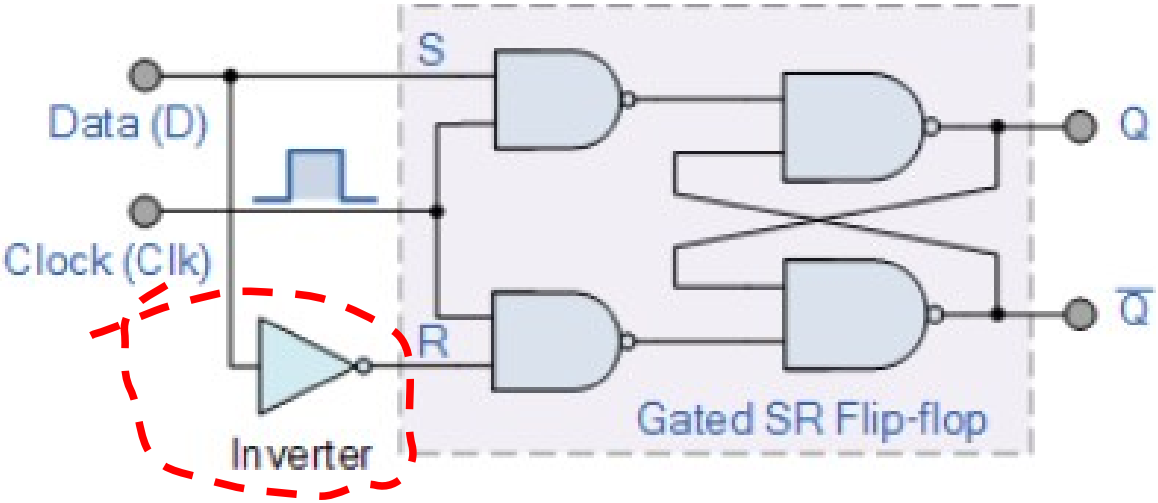


Flip-flop JK Mestre-Escravo

Não possui a condição proibida

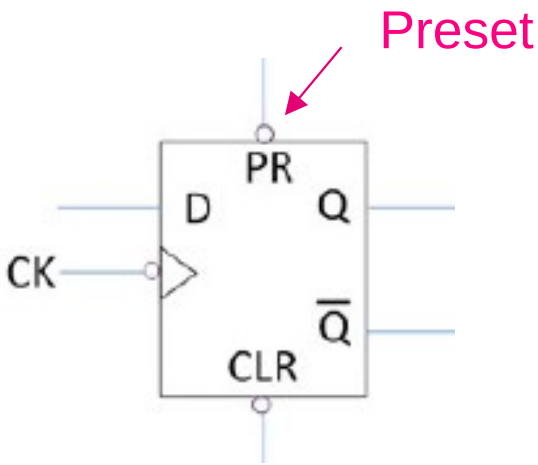


Flip-flop tipo D



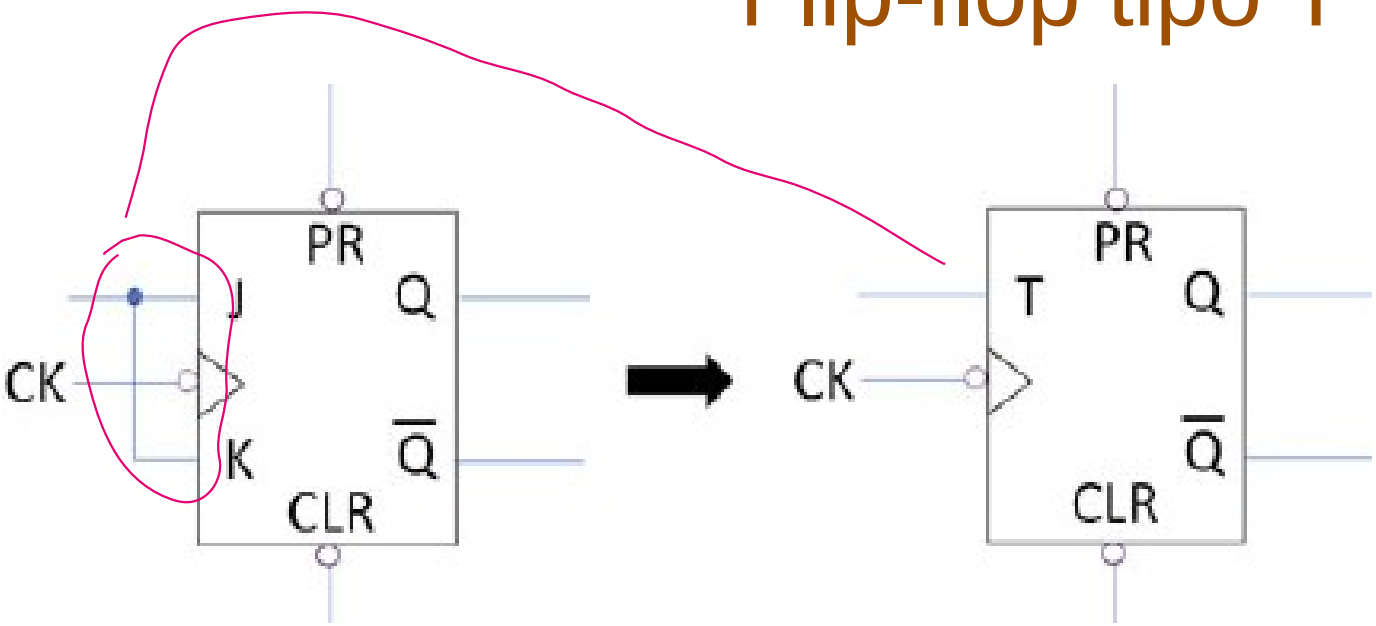
É um flip-flop SR com a adição de um inversor para evitar que as entradas S e R estejam no mesmo nível lógico

Flip-flop D			
J	K	D	Q_f
0	0	Impossível	/
0	1	0	0
1	0	1	1
1	1	Impossível	/



(outra representação)

Flip-flop tipo T



É uma variação do flip-flop JK, com clock

J e K sempre têm valores iguais

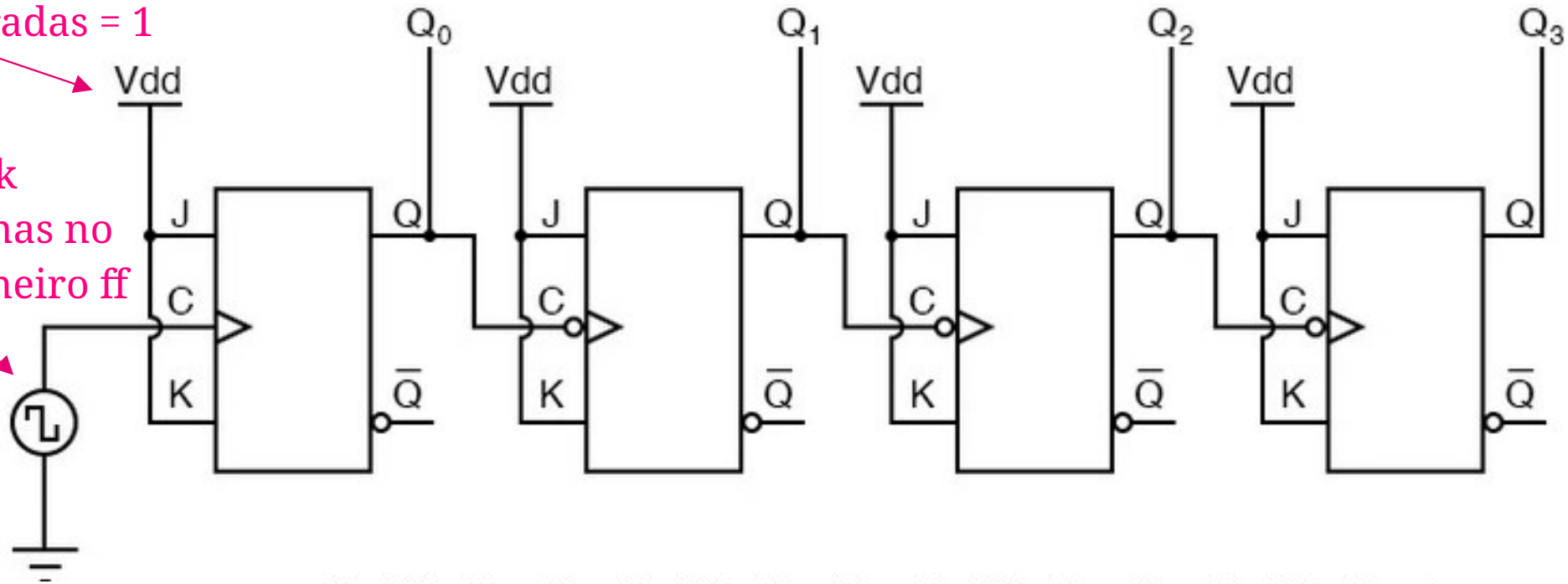
Usado em contadores assíncronos

Flip-flop T			
J	K	T	Q_f
0	0	0	Q_a
0	1	Impossível	/
1	0	Impossível	/
1	1	1	$\overline{Q_a}$

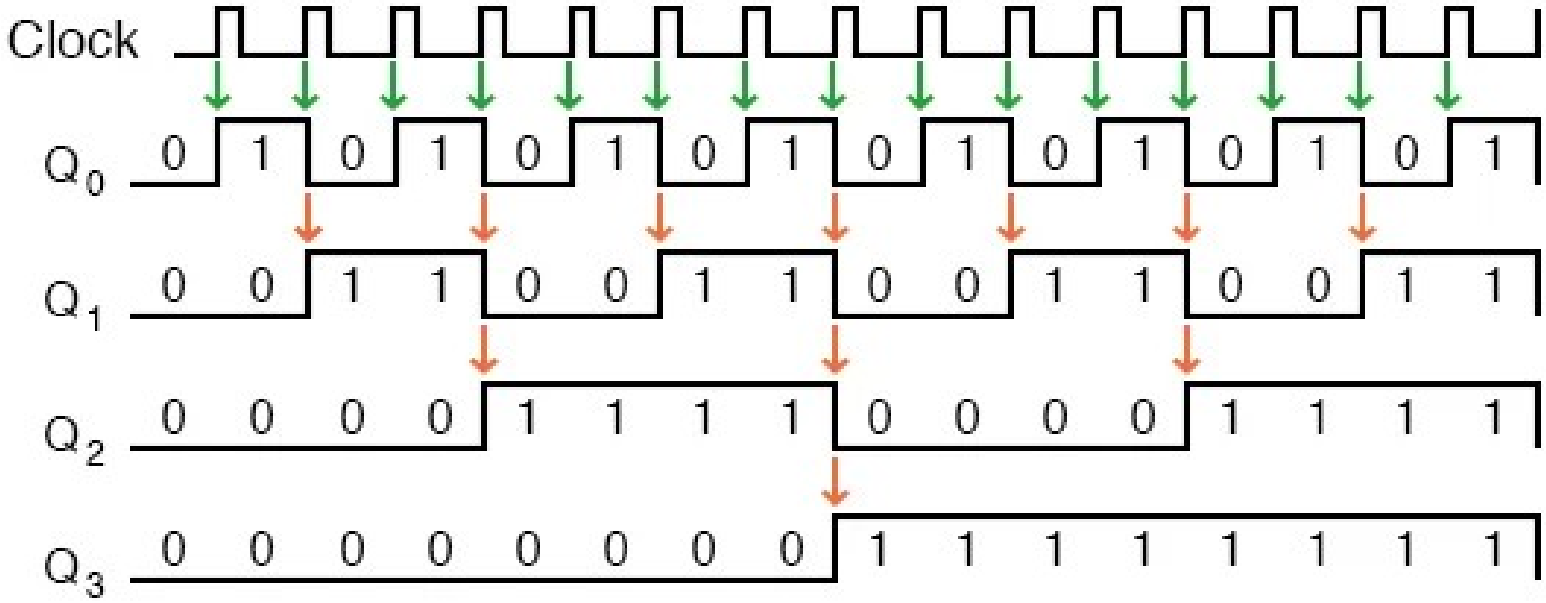
Contador assíncrono 4 bits progressivo

entradas = 1

clock apenas no primeiro ff



sensível a borda de subida



https://www.allaboutcircuits.com/textbook/digital/chpt-11/asynchronous-counters/

Contador síncrono de 4 bits progressivo

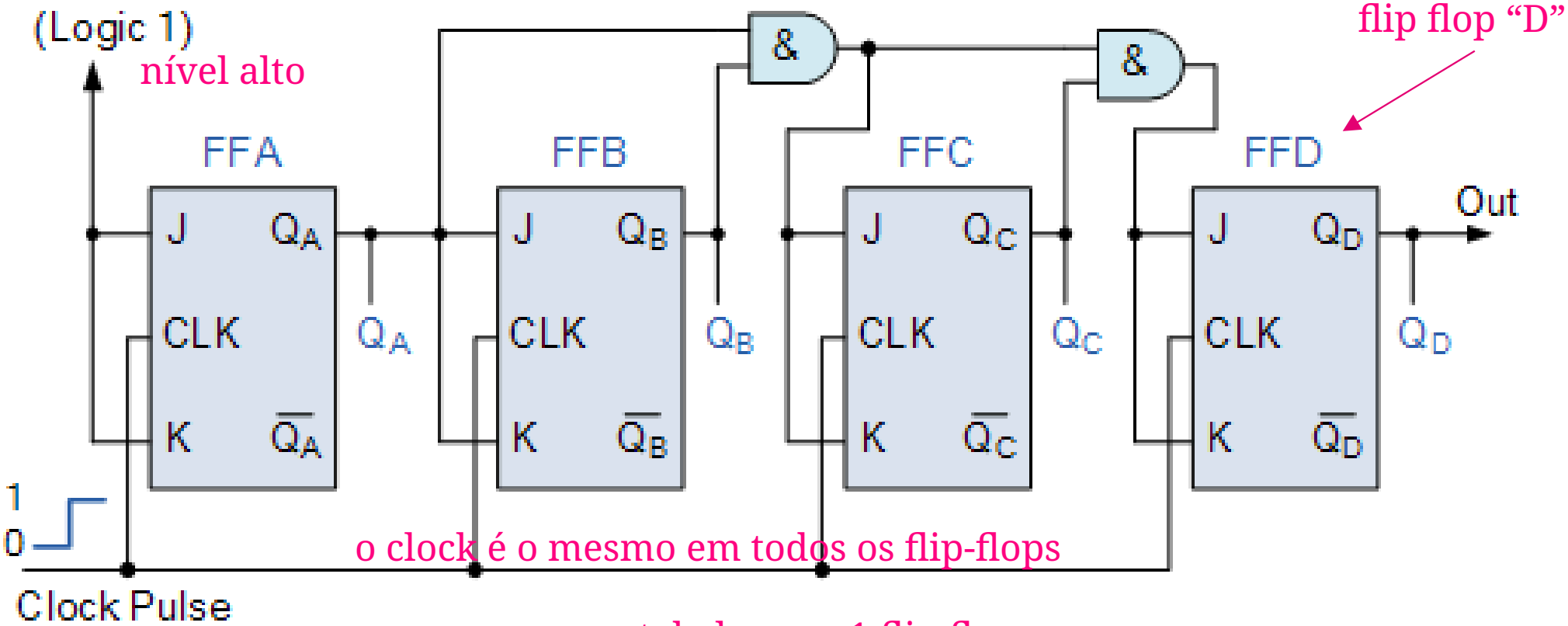


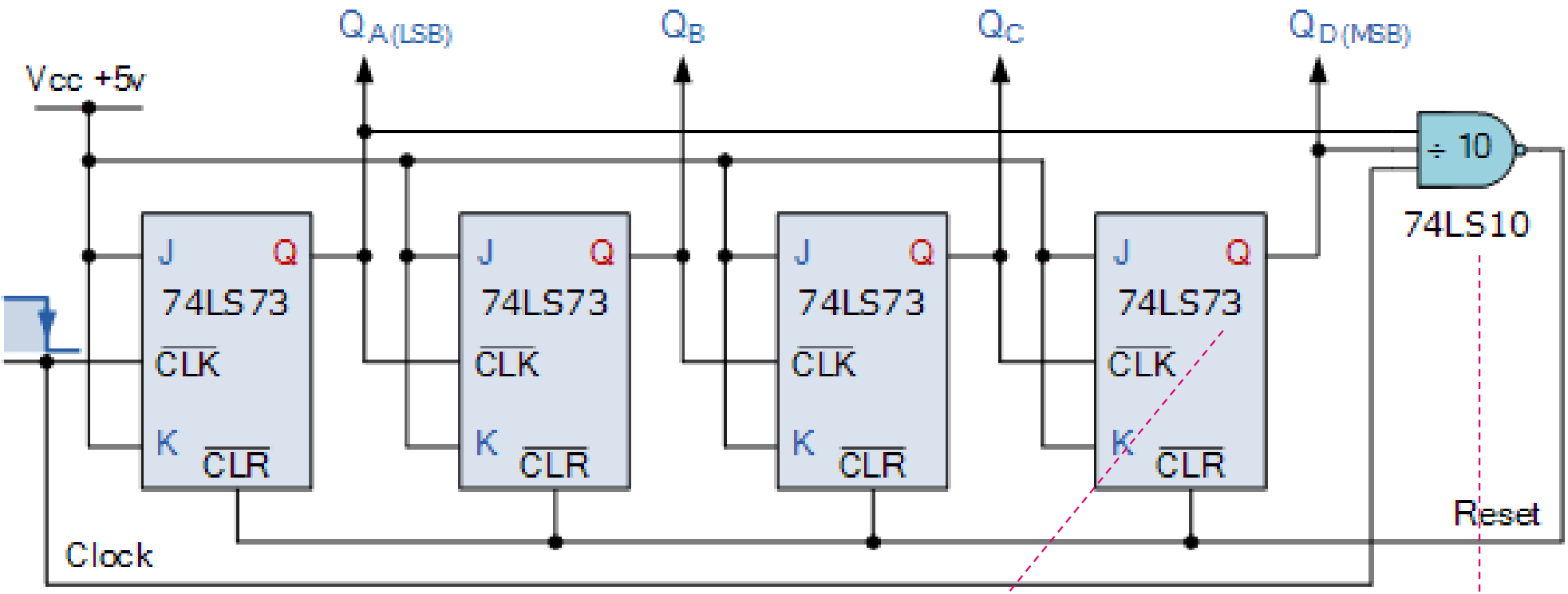
tabela para 1 flip flop

Q_a	Q_f	J	K
0	0 não muda	0	X
0	1	1	X
1	0	X	1
1	1 não muda	X	0

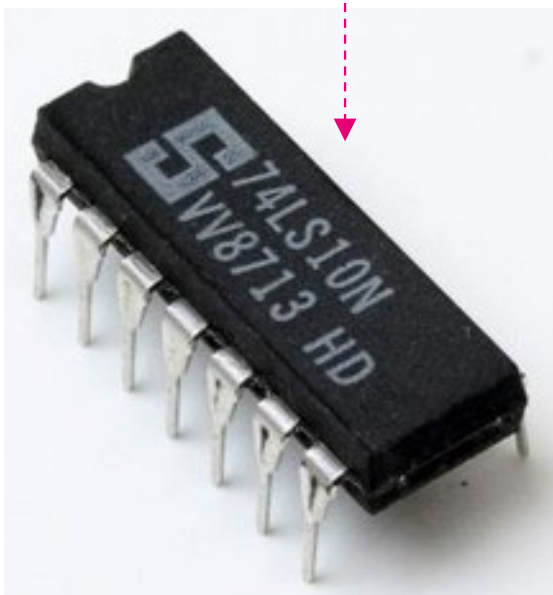
estado anterior

https://www.electronics-tutorials.ws/counter/count_3.html

Contador de década assíncrono



reseta para 0 quando a saída atinge 10_{10}, ou seja DCBA=1010_2

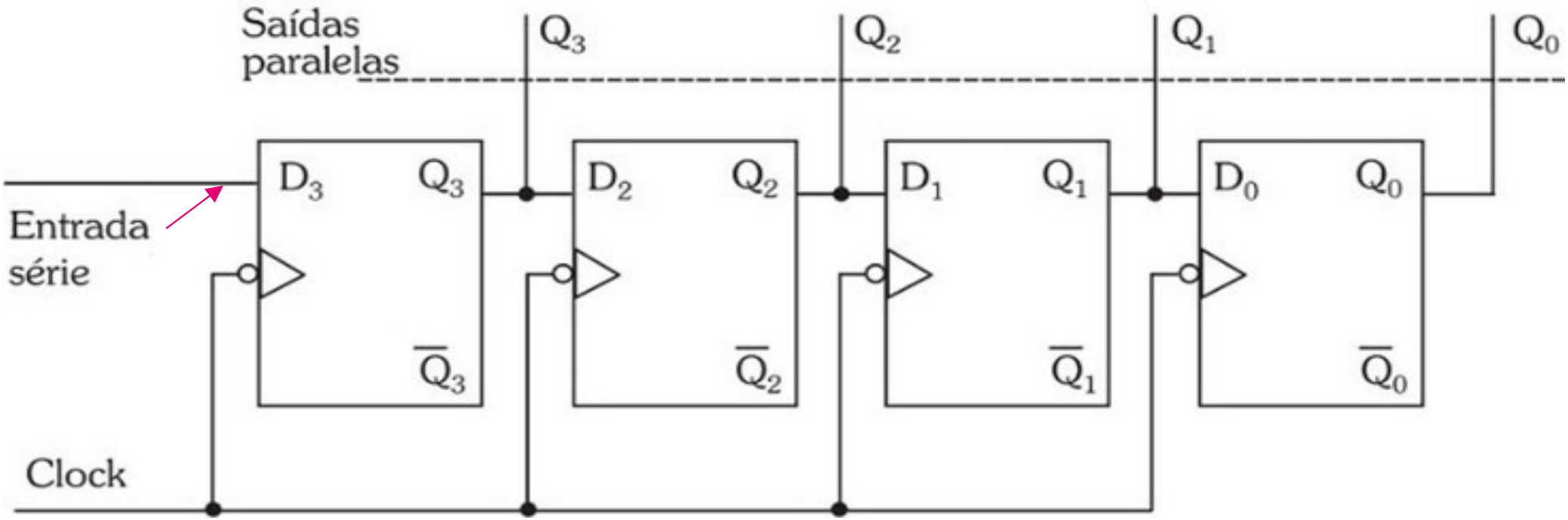


https://www.electronics-tutorials.ws/counter/count_2.html

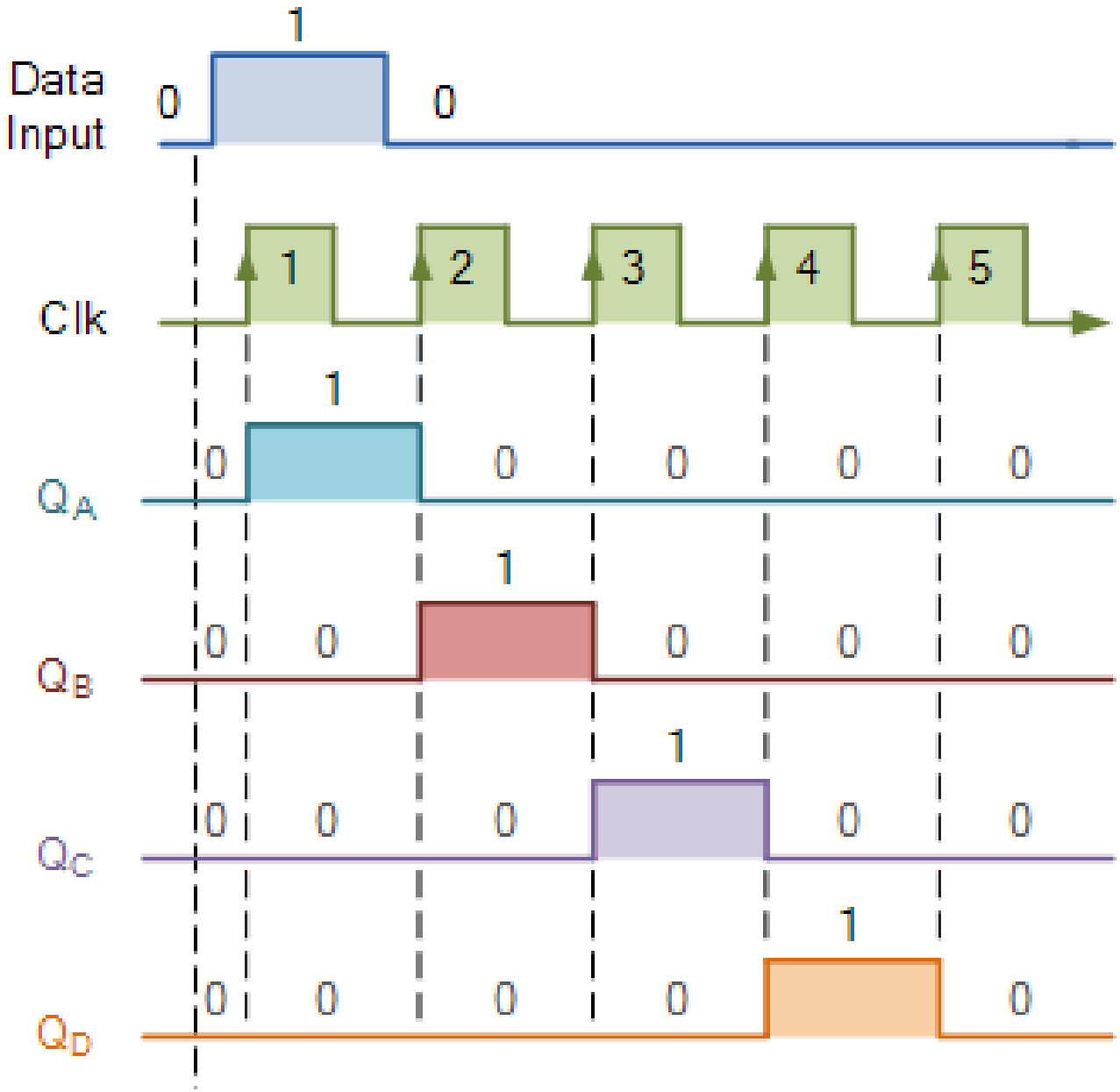
Registrador de deslocamento

Armazena mais de um bit

Conversão série-paralelo

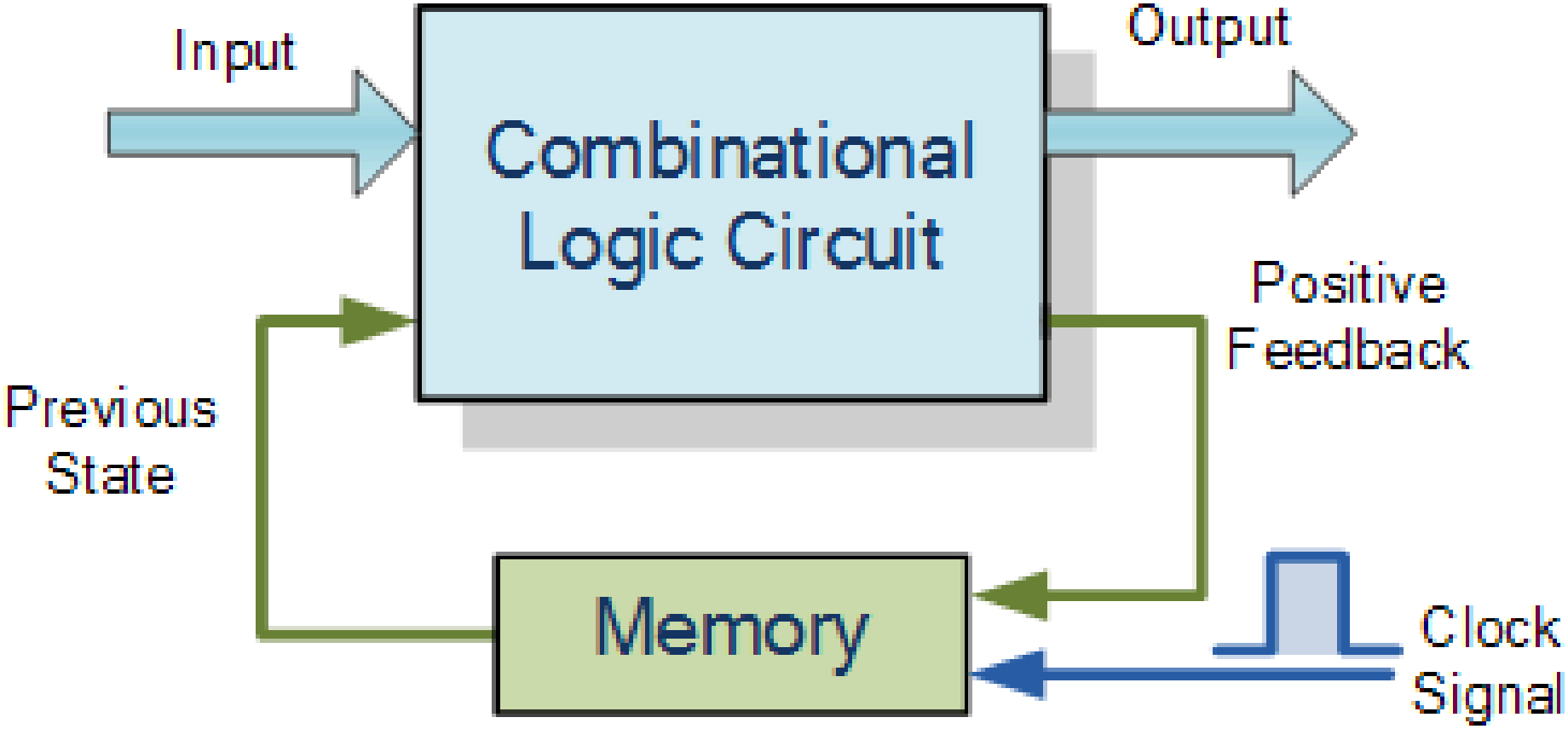


Registrador de deslocamento



https://www.electronics-tutorials.ws/sequential/seq_5.html

Sistemas sequenciais



https://www.electronics-tutorials.ws/sequential/seq_1.html

Sistemas sequenciais

Também chamados de máquinas de estados finitos

- Nº finito de estados possíveis de saída

Sequencia um conjunto de estados predeterminados

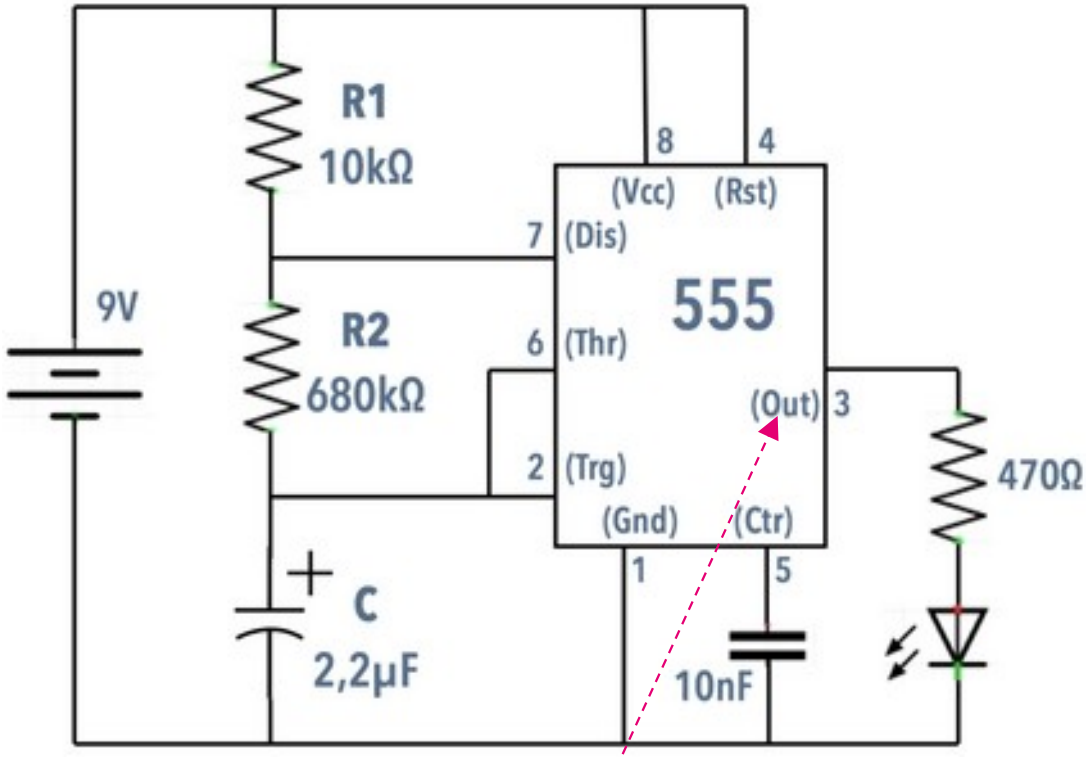
- Controlados por um clock
- E por outros sinais de entrada

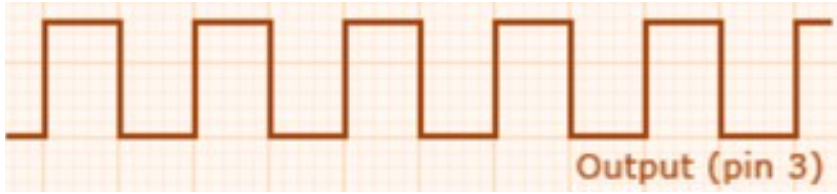
Dois modelos normalmente utilizados para implementar

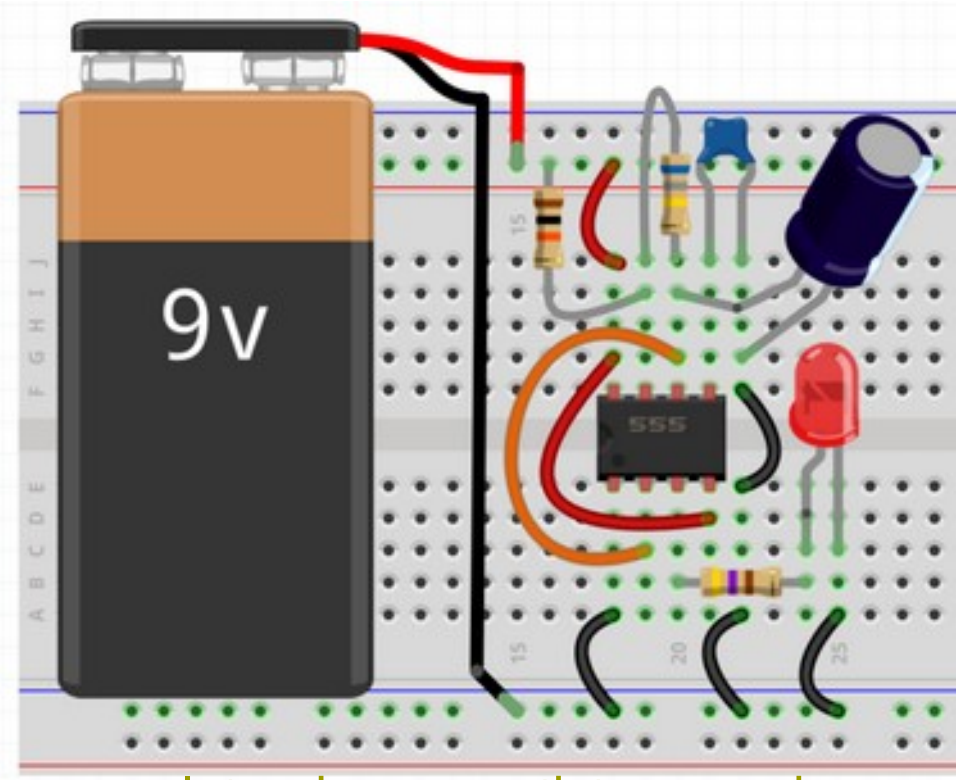
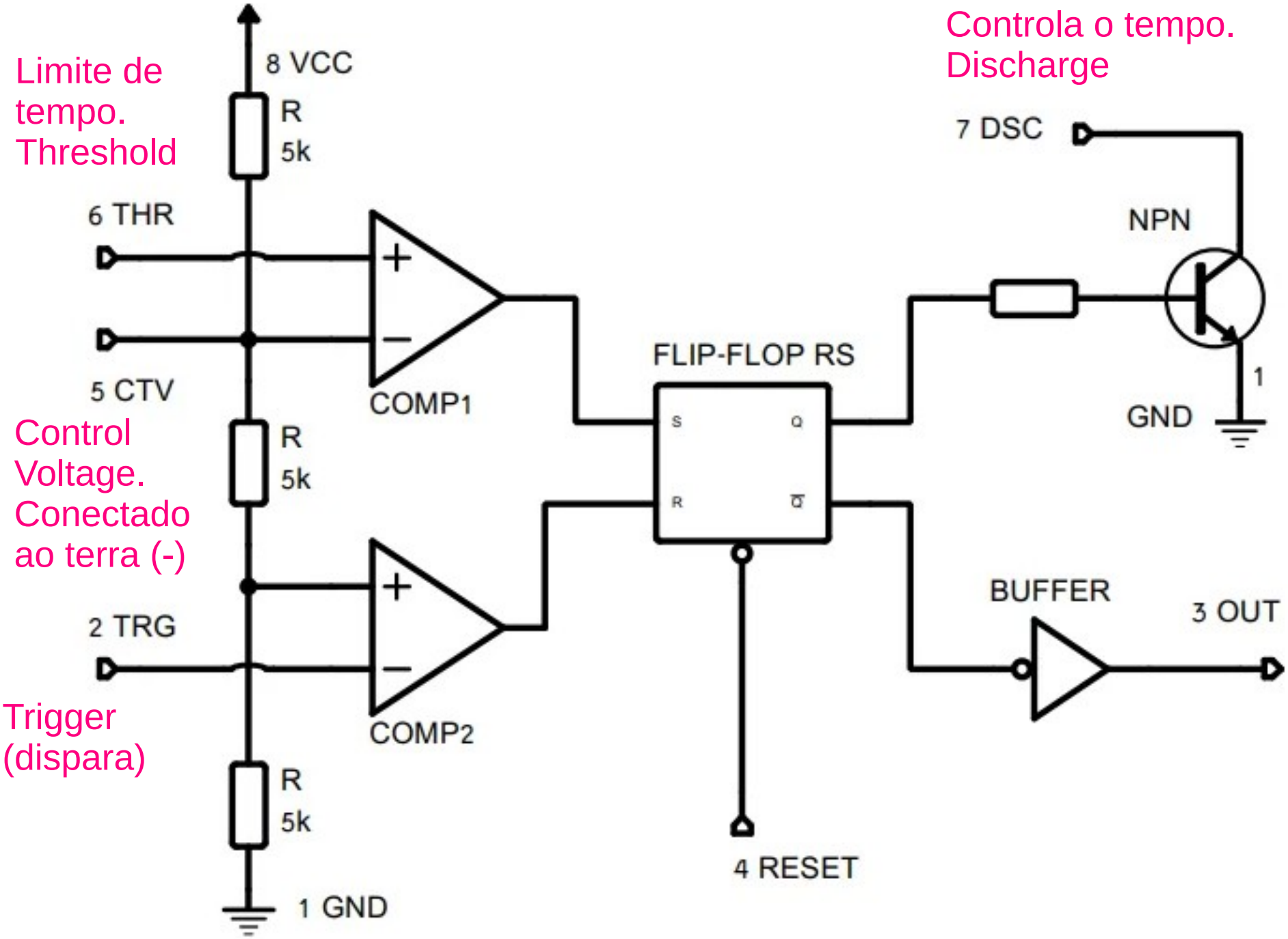
- Modelo de Moore
  - Síncrono
- Modelo de Mealy
  - Saídas podem atuar assincronamente

Temporizador 555

Pisca-pisca com LED
(modo de operação astável)







555 internamente